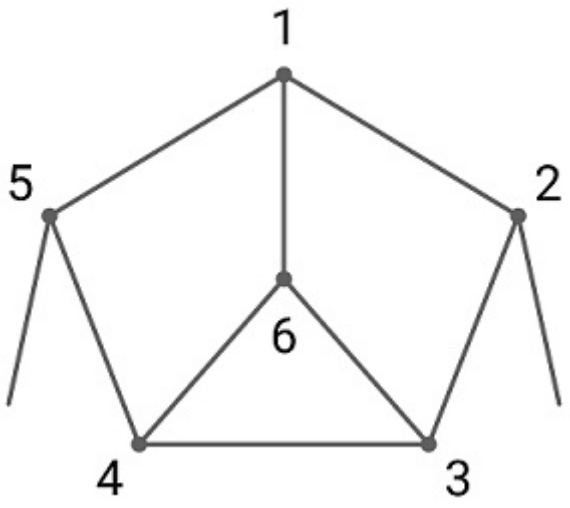

Enter a comma separated list of junction labels, or enter NIL.
NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEOUS CHARACTERS.

Answer format: Y9,T9,W9,L9

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

NIL

Deep Learning

Section Id :	64065375574
Section Number :	4
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	19
Number of Questions to be attempted :	19
Section Marks :	50
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0

Sub-Section Number : 1
Sub-Section Id : 640653161186
Question Shuffling Allowed : No

Question Number : 82 Question Id : 6406531045500 Question Type : MCQ
Correct Marks : 0
Question Label : Multiple Choice Question
THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : DEEP LEARNING (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)
Options :
6406533529613. ✔ YES
6406533529614. ✖ NO

Question Number : 83 Question Id : 6406531045501 Question Type : MCQ
Correct Marks : 0
Question Label : Multiple Choice Question
Note: If the base of log is not mentioned, use log base e.
Options :
6406533529615. ✔ Instructions has been mentioned above.
6406533529616. ✖ This Instruction is just for a reference & not for an evaluation.

Sub-Section Number : 2
Sub-Section Id : 640653161187
Question Shuffling Allowed : Yes

Question Number : 84 Question Id : 6406531045502 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Which of the following threshold θ values for an MP neuron implements the AND Boolean function, denoted as $f(\mathbf{x})$? Assume that the number of inputs x_i to the neuron is five and the neuron does not have any inhibitory inputs.

$$f(\mathbf{x}) = \begin{cases} 1, & \text{if } \sum_{i=1}^{5} x_i \geq \theta \\ 0, & otherwise \end{cases}$$

Options :
6406533529617. ✔ 5

6406533529618. ✖ 3

6406533529619. ✖ 1

6406533529620. ✖ 0

Question Number : 85 Question Id : 6406531045511 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

You are given a dataset of 10×10 *grayscale* images. Your goal is to build a 5-class classifier. You have to adopt one of the following two options:

- Model A: the input is flattened into a 100-dimensional vector, followed by a fully-connected layer with 5 neurons without bias
- Model B: the input is directly given to a convolutional layer with five 10×10 filters

Suppose you make your choice on the basis of the number of parameters in the models, p_A = number of parameters in ModelA, and similarly let p_B = number of parameters in ModelB.

Options :

6406533529643. ✖ $p_A > p_B$

6406533529644. ✔ $p_A = p_B$

6406533529645. ✖ $p_A < p_B$

6406533529646. ✖ None of these

Question Number : 86 Question Id : 6406531045519 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

In the context of RNNs, what structural feature of LSTMs helps reduce the impact of vanishing gradients?

Options :

6406533529656. ✖ Skip connections

6406533529657. ✔ Gated mechanisms like forget and input gates

6406533529658. ✖ Weight sharing across layers

6406533529659. ✖ Use of dropout

Sub-Section Number :	3
Sub-Section Id :	640653161188
Question Shuffling Allowed :	Yes

Question Number : 87 Question Id : 6406531045503 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

As per gradient descent, we should move towards 180 degrees with respect to gradient direction. What will happen if we move between 90 and 180 degrees? Consider the loss function to be convex.

Options :

6406533529621. ✖ The loss function will increase.

6406533529622. ✔ The loss function will decrease, although not to the maximum possible extent.

6406533529623. ✖ The loss function will remain the same.

6406533529624. ✖ Can't say. It depends on other parameters of the convex function.

Question Number : 88 Question Id : 6406531045508 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following statements about the parameter β in RMSProp is correct?

Options :

6406533529628. ✖ A higher value of β means that past gradients retain more influence on the moving average.

6406533529629. ✖ A lower value of β means past gradients have a lesser influence on the moving average.

6406533529630. ✖ A higher value of β means the current gradient will have lesser influence than past gradients.

6406533529631. ✖ A lower value of β means the current gradient will have higher influence than past gradients.

6406533529632. ✔ All of these

Question Number : 89 Question Id : 6406531045510 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Consider a feed-forward neural network containing three inputs and one output. It consists of 50 hidden layers, each with two neurons. The activation function used in all the hidden layers is ReLU, the output layer uses the Sigmoid activation function for binary classification. The network is trained with a binary cross-entropy loss function.
The network architecture is as follows:

- Input layer: 3 nodes
- Hidden layer: 2 nodes each
- Output layer: 1 node

Assume all weights are initialized to 1 and all biases to 0. For an input of [-10, 5, -20], what will be the output of the neural network?

Options :

6406533529637. ✖ 0

6406533529638. ✖ 1

6406533529639. ✔ 0.5

6406533529640. ✖ 0.25

6406533529641. ✖ -25

6406533529642. ✖ Insufficient information

Question Number : 90 Question Id : 6406531045520 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Given the attention weights $\alpha_{t,1} = 0.2$, $\alpha_{t,2} = 0.2$, $\alpha_{t,3} = 0.6$ and the corresponding encoder hidden states $h_1 = [1, 1, 1]$, $h_2 = [0, 1, 0]$, $h_3 = [0, 0, 1]$, calculate the context vector c_t.

Options :

6406533529660. ✖ [0.7, 0.3, 0.6]

6406533529661. ✖ [0.4, 0.2, 0.8]

6406533529662. ✖ [0.7, 0.6, 0.3]

6406533529663. ✔ [0.2, 0.4, 0.8]

Question Number : 91 Question Id : 6406531045522 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

In the transformer model, how are self-attention weights calculated for a given token?

Options :

6406533529665. ✖ By passing the token's embeddings to softmax function.

6406533529666. ✖ Using a feedforward neural network applied to the token's embeddings.

6406533529667. ✖ Through a convolutional operation applied to the token's neighborhood.

6406533529668. ✔ As the dot product between the token's embeddings and those of all other tokens, followed by a softmax operation.

Sub-Section Number :	4
Sub-Section Id :	640653161189
Question Shuffling Allowed :	No

Question Id : 6406531045504 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (92 to 94)

Question Label : Comprehension

Consider a sigmoid neuron that takes in an input vector $\mathbf{x} = \begin{bmatrix} 1 \\ 0.5 \\ 1 \end{bmatrix}$. The weight vector $\mathbf{w}$ is initialized to $\begin{bmatrix} 0.5 \\ 1 \\ 0.5 \end{bmatrix}$ and $\mathbf{b} = 1$. The output from the sigmoid neuron is

$$z = \mathbf{w}^{\mathbf{T}}\mathbf{x} + \mathbf{b}$$

$$\hat{\mathrm{y}} = \frac{1}{1 + exp^{-(z)}}$$

Suppose we use the following loss function

$$\mathcal{L} = \frac{1}{2}(y - \hat{y})^2$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 92 Question Id : 6406531045505 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the value of z
(write correct upto 2 decimals).

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

2.47 to 2.53

Question Number : 93 Question Id : 6406531045506 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the value of $\hat{\mathrm{y}}$
(write correct upto 2 decimals).

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.89 to 0.95

Question Number : 94 Question Id : 6406531045507 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

Update the weight vector once by running the vanilla Gradient Descent algorithm with $\eta = 2$. Assume the true label is $\mathbf{y} = \mathbf{0}$. What is the new loss value (that is, the loss computed after updating the weights and biases)? (consider upto two digits after the decimal for all the calculations)

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.36 to 0.42

Sub-Section Number :	5
Sub-Section Id :	640653161190
Question Shuffling Allowed :	Yes

Question Number : 95 Question Id : 6406531045509 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Which of the following techniques does NOT help prevent a model from overfitting?

Options :

6406533529633. ✖ Data augmentation

6406533529634. ✖ Dropout

6406533529635. ✖ Early stopping

6406533529636. ✔ None of these

Sub-Section Number :	6
Sub-Section Id :	640653161191
Question Shuffling Allowed :	Yes

Question Number : 96 Question Id : 6406531045512 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Suppose you have a vocabulary of 5,000 unique words and you want to train a CBOW model with a window size of 3 (on each side) and with an embedding dimension of 100. How many parameters (weights) will the embedding layer have?

Response Type : Numeric

Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
500000

Question Number : 97 Question Id : 6406531045513 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question
Assume that your CBOW model outputs a probability distribution over a vocabulary of 20,000 words for a given context. If the correct target word is word number 150, and the model's predicted probability for this word is 0.2, calculate the cross-entropy loss for this prediction. (use natural log)
Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
1.57 to 1.63

Question Number : 98 Question Id : 6406531045514 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question
In a Skip-gram model with a window size of 2 (on each side), how many **unique** pairs of target and context words will be generated for the following sentence:
'**Predicting the future is not magic, it is artificial intelligence**'

Note: Ignore the punctuations in your calculation
Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
32

Question Number : 99 Question Id : 6406531045521 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

An encoder RNN in a sequence-to-sequence model has the following specifications:

Embedding dimension = 11
Hidden state dimension = 7
Vocabulary size = 15
How many parameters does the encoder RNN have? (Assume no biases and assume we already have embeddings of all the input words).

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
126

Sub-Section Number :	7
Sub-Section Id :	640653161192
Question Shuffling Allowed :	No

Question Id : 6406531045515 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (100 to 102)
Question Label : Comprehension

Consider an RNN employing the following formulas to compute the state vector and output at time step t:

$$s_t = \sigma(Ux_i + Ws_{t-1} + b)$$
$$\hat{y}_t = O(Vs_t + c)$$

Here, σ and O denote the sigmoid and softmax functions, respectively. Assume that $s_t \in \mathbb{R}^2$, $\hat{y}_t \in \mathbb{R}^5$ and $x_i \in \mathbb{R}^6$

Based on the above data, answer the given subquestions.
Sub questions

Question Number : 100 Question Id : 6406531045516 Question Type : SA Correct Marks : 2
Question Label : Short Answer Question
With a total of 7 time steps (T = 7), implying prediction for a word of length 6), what is the total count of parameters (including bias) within the network?
Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Possible Answers :

33

Question Number : 101 Question Id : 6406531045517 Question Type : MCQ
Correct Marks : 3

Question Label : Multiple Choice Question

If all the parameters (including the bias) in the network are initialized to zero, what will be the predicted $\hat{y}_2$ at time step 2 (assuming indices start at 1) for the input $[0, 0, 1, 0, 0, 0]^T$?

Options :

6406533529651. ✖ $[0, 0.5, 0.5, 0, 0]^T$

6406533529652. ✖ $[0.1, 0.1, 0.6, 0.1, 0.1]^T$

6406533529653. ✖ $[0, 0, 1, 0, 0]^T$

6406533529654. ✔ $[0.2, 0.2, 0.2, 0.2, 0.2]^T$

Question Number : 102 Question Id : 6406531045518 Question Type : SA
Correct Marks : 2

Question Label : Short Answer Question

If all the parameters (including bias) in the network are initialized to zero, what will be the total loss after 10 time steps (assume that indices start with 1) for the input $[0, 0, 1, 0, 0, 0]^T$? The ground truth ($y \in \mathbb{R}^5$) for each time step is given by the following sequence $[y_0, y_4, y_1, y_2, y_3, y_4, y_4, y_4, y_2, y_3]$. Assume the loss to be cross-entropy at each time step. (Use natural log and write the answer correct up to two decimal places.) each y is one-hot encoder (i.e $y_0$means $= [1, 0, 0, 0, 0]^T$)

Response Type : Numeric

Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :

16 to 16.2

Sub-Section Number :	8
Sub-Section Id :	640653161193
Question Shuffling Allowed :	Yes

Question Number : 103 Question Id : 6406531045523 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question
In the context of the Transformer model's encoder-decoder architecture, which of the following statements are accurate?
Options :
6406533529669. ✔ The encoder in a Transformer model is responsible for converting the input sequence into a continuous representation that the decoder can then use to generate the output sequence.
6406533529670. ✖ The decoder in a Transformer model only attends to the encoder's output without any form of self-attention on its own inputs.
6406533529671. ✔ Multi-head attention in the encoder allows the model to jointly attend to information from different representations at different positions.
6406533529672. ✔ The decoder's self-attention mechanism includes a masking component to prevent attending to future positions, ensuring the model generates outputs one step at a time.

Sub-Section Number :	9
Sub-Section Id :	640653161194
Question Shuffling Allowed :	Yes

Question Number : 104 Question Id : 6406531045524 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question
Consider a Transformer model with the following specifications for the decoder part:

• Input dimension (embedding size): 16
• Number of heads in multi-head attention: 2
• Head output dimension: 8
• Dimension of the feed-forward network: 32
• Number of layers in the decoder: 2
Note: The final output from the attention heads is obtained by concatenating the outputs from the individual heads and hence does not cost any additional parameters.
Assume that each decoder layer contains:
• One multi-head attention mechanism for self-attention.